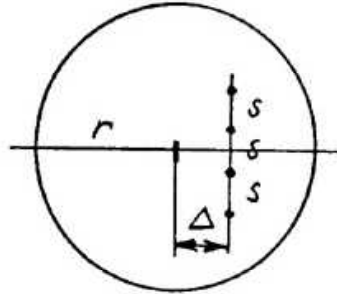


I.3) Probe Array Perpendicular to a Diameter.thickness $t < \frac{s}{2}$

Swartzendruber (j) published the geometric factor for this geometri:

$$\rho = G \frac{V}{I}, \quad G = \frac{\pi}{\ln 2} \cdot t \cdot C_2\left(\frac{s}{d}, \frac{\Delta}{d}\right) \quad (22)$$

where:

$$C_2\left(\frac{s}{d}, \frac{\Delta}{d}\right) = \frac{1}{1 + \frac{1}{2\ln 2} \ln \frac{\alpha_1 \alpha_2}{4 \alpha_3 \alpha_4}} \quad (23)$$

$$\alpha_1 = (v_2 - v_1)^2 + (u_2 + u_1)^2$$

$$\alpha_2 = (v_2 + v_1)^2 + (u_2 + u_1)^2$$

$$\alpha_3 = (v_2 - v_1)^2 + (u_2 - u_1)^2$$

$$\alpha_4 = (v_2 + v_1)^2 + (u_2 - u_1)^2$$

$$u_1 = \frac{3\frac{s}{r}}{\left(1 + \frac{\Delta}{r}\right)^2 + \frac{9}{4}\left(\frac{s}{r}\right)^2}$$

$$u_2 = \frac{\frac{s}{r}}{\left(1 + \frac{\Delta}{r}\right)^2 + \frac{1}{4}\left(\frac{s}{r}\right)^2}$$

$$v_1 = \frac{1 - \left(\frac{\Delta}{r}\right)^2 - \frac{9}{4}\left(\frac{s}{r}\right)^2}{\left(1 + \frac{\Delta}{r}\right)^2 + \frac{9}{4}\left(\frac{s}{r}\right)^2}$$

$$v_2 = \frac{1 - \left(\frac{\Delta}{r}\right)^2 - \frac{1}{4}\left(\frac{s}{r}\right)^2}{\left(1 + \frac{\Delta}{r}\right)^2 + \frac{1}{4}\left(\frac{s}{r}\right)^2}$$